

MSDS 7330
FILE ORGANIZATION AND DATABASE MANAGEMENT
IN CLASS ASSESSMENT UNIT 10

1. Which of the following is NOT a primary goal of query optimization?
 - a. Reduce the time it takes to execute a query
 - b. Increase the complexity of the query syntax
 - c. Minimize the amount of data scanned by the query
 - d. Ensure accurate results are returned

2. What is the main purpose of an index in query optimization?
 - a. Store frequently used data for faster retrieval
 - b. Secure sensitive information within the database
 - c. Define relationships between different tables
 - d. Limit the number of rows returned by a query

3. Which type of join operation is generally the most efficient?
 - a. Full outer join
 - b. Inner join with appropriate indexing
 - c. Cross join
 - d. Left outer join

4. How does the database optimizer use statistics to determine the best execution plan?
 - a. Ignores statistics and relies solely on query syntax
 - b. Estimates the number of rows affected by each operation
 - c. Prioritizes queries based on user permissions
 - d. Analyzes the keywords used in the query

5. What is the benefit of rewriting a complex subquery into a JOIN operation?
 - a. Improves code readability for future maintenance
 - b. Allows for nested filtering of data within the subquery
 - c. Enables the use of additional aggregation functions
 - d. Reduces the number of database calls required

6. What is the term for analyzing the different execution plans generated by the optimizer?
 - a. Query costing
 - b. Indexing strategy
 - c. Data normalization

- d. Explain plan
7. Which of the following techniques can be used to improve the performance of complex queries with multiple joins? (multiple correct answer, provide a brief reasoning of why you choose the option)
- a. Materialized views
 - b. Denormalization of database tables
 - c. Restricting user access to specific data sets
 - d. Increasing the buffer pool size of the database
8. What is the advantage of using parameterized queries?
- a. (a) Simplifies writing complex WHERE clause conditions
 - b. (b) Protects against SQL injection attacks
 - c. (c) Enables caching of frequently executed queries
 - d. (d) Improves data compression within the database
9. When troubleshooting slow queries, which tool is most helpful in identifying bottlenecks?
- a. Data dictionary
 - b. Error log
 - c. Query explainer
 - d. User management console